MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC TEST – 01 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 12: Surface Areas and Volumes** | **MM:** 24

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets ([**0.5 Mark**], [**1 Mark**], [**1.5 Marks**], [**2 Marks**]). Alternative valid methods receive full credit.
3. Total marks of printed questions Q1–Q11 $= 5 + 4 + 6 + 5 + 4 = 24$ Marks (matching section totals; source paper header prints MM: 30).

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

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Q1. A hollow spherical shell of internal and external diameters 4 cm and 8 cm is melted into a cone of base diameter 8 cm. The height of the cone is: [1]

(A) 12 cm (B) 14 cm (C) 15 cm (D) 18 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 14 cm** [0.5 Mark]

Given: Shell: $r_1 = \frac{4}{2} = 2$ cm, $r_2 = \frac{8}{2} = 4$ cm. Cone: $R = \frac{8}{2} = 4$ cm.

Formula / Theorem Used: $\frac{1}{3}\pi R^2 h = \frac{4}{3}\pi(r_2^3 - r_1^3)$

$$4^2 h = 4(4^3 - 2^3) \implies 16h = 4(64 - 8) = 4 \times 56 \implies h = \frac{224}{16} = \boxed{\mathbf{14}\text{ cm}}$$

Q2. A solid piece of iron of dimensions 49 cm $\times$ 33 cm $\times$ 24 cm is moulded into a solid sphere. The radius of the sphere is: [1]

(A) 21 cm (B) 28 cm (C) 35 cm (D) 14 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) 21 cm** [0.5 Mark]

Given: Cuboid $49 \times 33 \times 24$ cm^3 moulded into sphere of radius r.

Formula / Theorem Used: $\frac{4}{3}\pi r^3 = lbh$

$$\frac{4}{3} \times \frac{22}{7} \times r^3 = 49 \times 33 \times 24 \implies r^3 = \frac{49 \times 33 \times 24 \times 21}{4 \times 22} = 49 \times 3 \times 3 \times 7 = 7^3 \times 3^3 \implies \boxed{\mathbf{r = 21}\text{ cm}}$$

Q3. The ratio of lateral surface area to the total surface area of a cylinder with base diameter 1.6 m and height 20 cm is: [1]

(A) 1 : 5 (B) 1 : 7 (C) 1 : 8 (D) 1 : 9

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) 1 : 5** [0.5 Mark]

Given: Diameter $= 1.6$ m $= 160$ cm $\implies r = 80$ cm, $h = 20$ cm.

Formula / Theorem Used: $\dfrac{\text{LSA}}{\text{TSA}} = \dfrac{2\pi rh}{2\pi r(h+r)} = \dfrac{h}{h+r}$

$$\frac{\text{LSA}}{\text{TSA}} = \frac{20}{20+80} = \frac{20}{100} = \boxed{\mathbf{1:5}}$$

Q4. A solid consists of a circular cylinder surmounted by a right circular cone of height h. If the total volume of the solid is 3 times the volume of the cone, the height of the cylinder is: [1]

(A) h (B) $\dfrac{2h}{3}$ (C) $\dfrac{3h}{2}$ (D) $2h$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B)** $\dfrac{2h}{3}$ [0.5 Mark]

Given: Equal base radius r. Total volume $V_{\text{cyl}} + V_{\text{cone}} = 3V_{\text{cone}} \implies V_{\text{cyl}} = 2V_{\text{cone}}$.

Formula / Theorem Used: $\pi r^2 H = 2\left(\dfrac{1}{3}\pi r^2 h\right) \implies \boxed{\mathbf{H = \dfrac{2h}{3}}}$

Q5. **Assertion (A):** Two solid metallic right circular cones of equal height and base radii 3 cm and 4 cm are melted and recast into a solid sphere of radius 5 cm. The cones are of height 20 cm each.
Reason (R): If two right circular cones of height h and base radii r_1, r_2 are recast into a sphere of radius R, then $h = \dfrac{4R^3}{r_1^2 + r_2^2}$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** [0.5 Mark]

Verification: Equating volumes: $\frac{1}{3}\pi h(r_1^2 + r_2^2) = \frac{4}{3}\pi R^3 \implies h = \frac{4R^3}{r_1^2+r_2^2}$ (Reason (R) is true).

For $r_1 = 3, r_2 = 4, R = 5$: $h = \frac{4(125)}{9+16} = \frac{500}{25} = 20$ cm. Both (A) and (R) are true, and (R) explains (A).

Q6. The radii of the bases of a cylinder and a cone are in the ratio 3 : 4 and their heights are in the ratio 2 : 3. What is the ratio of their volumes? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Ratio Setup [1 Mark]

Let $\frac{r_1}{r_2} = \frac{3}{4}$ and $\frac{h_1}{h_2} = \frac{2}{3}$.

Formula / Theorem Used: $\frac{V_{\text{cylinder}}}{V_{\text{cone}}} = \frac{\pi r_1^2 h_1}{\frac{1}{3}\pi r_2^2 h_2} = 3\left(\frac{r_1}{r_2}\right)^2\left(\frac{h_1}{h_2}\right)$

Step 2: Substitution & Simplification [1 Mark]

$$\frac{V_{\text{cylinder}}}{V_{\text{cone}}} = 3 \times \left(\frac{3}{4}\right)^2 \times \frac{2}{3} = 3 \times \frac{9}{16} \times \frac{2}{3} = \frac{9}{8} \implies \boxed{\mathbf{9:8}}$$

Q7. If the heights of two right circular cones are in the ratio 1 : 2 and the perimeters of their bases are in the ratio 3 : 4, what is the ratio of their volumes? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Finding Radius Ratio from Perimeter Ratio [1 Mark]

Given $\frac{h_1}{h_2} = \frac{1}{2}$ and $\frac{2\pi r_1}{2\pi r_2} = \frac{3}{4} \implies \frac{r_1}{r_2} = \frac{3}{4}$.

Formula / Theorem Used: $\frac{V_1}{V_2} = \frac{\frac{1}{3}\pi r_1^2 h_1}{\frac{1}{3}\pi r_2^2 h_2} = \left(\frac{r_1}{r_2}\right)^2\left(\frac{h_1}{h_2}\right)$

Step 2: Arithmetic [1 Mark]

$$\frac{V_1}{V_2} = \left(\frac{3}{4}\right)^2 \times \frac{1}{2} = \frac{9}{16} \times \frac{1}{2} = \frac{9}{32} \implies \boxed{\mathbf{9:32}}$$

Q8. Two identical solid cubes of side a are joined end to end. Find the total surface area of the resulting cuboid. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions of Resulting Cuboid [1 Mark]

When two cubes of edge a are joined end-to-end, length doubles while breadth and height stay a:
Length $L = a + a = 2a$, Breadth $B = a$, Height $H = a$.

Formula / Theorem Used: Total Surface Area of Cuboid $= 2(LB + BH + HL)$

Step 2: Substitution & Simplification [2 Marks]

$$\text{TSA} = 2((2a)(a) + (a)(a) + (a)(2a)) = 2(2a^2 + a^2 + 2a^2) = 2(5a^2) = \boxed{\mathbf{10a^2}\text{ sq. units}}$$

Q9. An empty cone is of radius 3 cm and height 12 cm. Ice-cream is filled in it so that lower part which is $\frac{1}{6}$ of the volume is unfilled, but a hemispherical scoop of radius 3 cm is on top. Find the volume of ice-cream. (Take $\pi = 3.14$) **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions & Volume Formula [1 Mark]

Radius $r = 3$ cm, cone height $h = 12$ cm. Since $\frac{1}{6}$ of cone is unfilled, filled fraction $= 1 - \frac{1}{6} = \frac{5}{6}$.

Formula / Theorem Used: $V_{\text{ice-cream}} = \frac{5}{6}V_{\text{cone}} + V_{\text{hemi}} = \frac{5}{6}\left(\frac{1}{3}\pi r^2 h\right) + \frac{2}{3}\pi r^3$

Step 2: Factoring π & Arithmetic ($\pi = 3.14$) [2 Marks]

$$V = \frac{5}{6}\left(\frac{1}{3}\pi \times 9 \times 12\right) + \frac{2}{3}\pi(27) = \frac{5}{6}(36\pi) + 18\pi = 30\pi + 18\pi = 48\pi$$

$$V = 48 \times 3.14 = \boxed{\mathbf{150.72}\ \text{cm}^3}$$

SECTION D: Long Answer / HOTS Question

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Q10. A cistern, internally measuring 150 cm × 120 cm × 110 cm, has 129,600 cm^3 of water in it. Porous bricks of dimensions 22.5 cm × 7.5 cm × 6.5 cm are placed in the water. Each brick absorbs $\frac{1}{17}$th of its own volume of water. Find how many bricks can be put in the cistern without water overflowing to the brim. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Empty Space Available in Cistern [1.5 Marks]

Formula / Theorem Used: Effective Displacement per Brick $= \left(1 - \frac{1}{17}\right)V_{\text{brick}} = \frac{16}{17}V_{\text{brick}}$

Total volume of cistern $= 150 \times 120 \times 110 = 19,80,000$ cm^3.

Remaining capacity to brim $= 19,80,000 - 1,29,600 = 18,50,400$ cm^3.

Step 2: Net Volume Displaced per Brick [1.5 Marks]

Volume of 1 brick $= 22.5 \times 7.5 \times 6.5 = \frac{45}{2} \times \frac{15}{2} \times \frac{13}{2} = \frac{8775}{8} = 1096.875$ cm^3.

Suppose n bricks are placed: net displaced volume $= n \times \frac{16}{17} \times \frac{8775}{8} = n \times \frac{17550}{17}$ cm^3.

Step 3: Maximum Integer Number of Bricks [2 Marks]

$$n \times \frac{17550}{17} \leq 18,50,400 \implies n \leq \frac{1850400 \times 17}{17550} = \frac{69904}{39} \approx 1792.41$$

Adding 1793 bricks would overflow water, so take floor integer: $\boxed{\mathbf{1792}\ \text{bricks}}$.

Q11. Case Study: The Commonwealth Games Victory Podium

Organisers constructed a 3-tier wooden victory podium for medal ceremonies consisting of three joined rectangular cuboids placed side-by-side on the ground: 1st Position Block (Gold, Center) of size 40 cm × 50 cm × 40 cm, 2nd Position Block (Silver, Left) of size 40 cm × 50 cm × 24 cm, and 3rd Position Block (Bronze, Right) of size 40 cm × 50 cm × 12 cm. [4]

(i) Find the surface area of the front vertical face of the entire victory stand. [1]

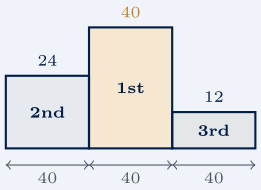
(ii) Find the volume of the block designated for the 2nd position athlete (Silver). [1]

(iii) Find the volume of the block designated for the 1st position winner (Gold). [1]

(iv) Find the volume of the block designated for the 3rd position athlete (Bronze). [1]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Extracted Scenario Data: Three cuboids of common width $w = 40$ cm, depth $d = 50$ cm, and heights $h_1 = 40$ cm (Gold), $h_2 = 24$ cm (Silver), $h_3 = 12$ cm (Bronze).



Formula / Theorem
Used: Rect. Area $= w \times h$ and Cuboid Volume $= l \times b \times h$

Sub-part (i): Front Vertical Face Area [1 Mark]

Area $= (40 \times 24) + (40 \times 40) + (40 \times 12) = 960 + 1600 + 480 =$ $\boxed{\mathbf{3040}\ \text{cm}^{\mathbf{2}}}$ (note: 3800 cm^2 if 50 cm faces front).

Sub-part (ii): Volume of 2nd Position Block (Silver) [1 Mark]

$V_{\text{Silver}} = 40 \times 50 \times 24 = 2000 \times 24 =$ $\boxed{\mathbf{48,000}\ \text{cm}^{\mathbf{3}}}$.

Sub-part (iii): Volume of 1st Position Block (Gold) [1 Mark]

$V_{\text{Gold}} = 40 \times 50 \times 40 = 2000 \times 40 =$ $\boxed{\mathbf{80,000}\ \text{cm}^{\mathbf{3}}}$.

Sub-part (iv): Volume of 3rd Position Block (Bronze) [1 Mark]

$V_{\text{Bronze}} = 40 \times 50 \times 12 = 2000 \times 12 =$ $\boxed{\mathbf{24,000}\ \text{cm}^{\mathbf{3}}}$.

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